

Worksheet 8B: Probability Distributions

Series 8: Probability Deepened
AIML Foundations Mathematics
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What This Worksheet Is About

*A probability distribution assigns a probability to every possible outcome. Some distributions appear again and again across science and AI. The **Binomial** counts successes in repeated trials. The **Poisson** counts rare events. The **Normal** describes noise, measurement error, and — by a remarkable theorem — the average of almost anything. Knowing these distributions means you can recognise the shape of a problem and reach immediately for the right tool.*

Part A: Bernoulli and Binomial Distributions (12 problems)

Bernoulli: one trial, $P(X = 1) = p$, $P(X = 0) = 1 - p$; mean = p , variance = $p(1 - p)$.

Binomial(n, p): k successes in n independent trials:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad \binom{n}{k} = \frac{n!}{k!(n - k)!}$$

Mean = np , Variance = $np(1 - p)$.

1. Biased coin: $P(H) = 0.6$. (a) $P(X = 3)$ in 5 flips. (b) Table: $P(X = k)$ for $k = 0, \dots, 5$; verify sum = 1. (c) Mean and variance.
2. ML model: 80% accuracy, 10 images. (a) $P(\text{exactly 8 correct})$. (b) $P(9 \text{ or } 10 \text{ correct})$. (c) Expected correct and standard deviation.
3. Show that $\sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} = 1$ using $(p + (1 - p))^n = 1$.
4. Compute: (a) $\binom{5}{2}$ (b) $\binom{6}{0}$ (c) $\binom{10}{3}$ (d) $\binom{8}{4}$
5. Quality control: 20 items, 10% defective. (a) State distribution. (b) $P(X = 0)$. (c) $P(X \leq 2)$. (d) Expected defectives.
6. For Binomial(20, 0.5): (a) Mode? (b) Mean? (c) Shape as n increases?
7. $X \sim \text{Binomial}(100, 0.3)$: mean, variance, and standard deviation.
8. **A/B test.** Version B shown to 500 users; 15 clicks observed. (a) Expected clicks if rate = 2%? (b) $P(X = 15)$ under 2% rate (leave in formula form). (c) Does 15/500 suggest a different rate?
9. For the geometric distribution $P(X = k) = (1 - p)^{k-1}p$: a footballer scores penalties with $p = 0.25$. Find $P(X = 3)$, $P(X = 4)$, $P(X = 5)$.

10. $X \sim \text{Geometric}(p = 0.1)$: mean and variance?
11. If $X \sim \text{Binomial}(n = 200, p = 0.01)$, what Poisson distribution approximates X ?
12. True or false: (a) Binomial is symmetric when $p = 0.5$. (b) The sum of all Binomial probabilities is always 1. (c) Variance can exceed the mean for a Binomial. (d) $\text{Binomial}(1, p)$ is a Bernoulli.

Part B: Poisson Distribution (10 problems)

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots \quad \text{Mean} = \text{Variance} = \lambda$$

13. Website: 4 visits/minute. (a) $P(X = 4)$. (b) $P(X = 0)$. (c) $P(X > 6)$. (d) Variance?
14. Typos: 2 per page. (a) $P(X = 0)$. (b) $P(X = 2)$. (c) $P(X \leq 2)$. (d) $P(X > 3)$.
15. Hospital: 10 patients/hour. (a) $P(X = 8)$ (leave as expression; $e^{-10} \approx 0.0000454$). (b) Expected patients in 30 minutes. (c) Rate per minute?
16. Radiation: 0.5 impacts/second. (a) $P(X = 0)$, $P(X = 1)$, $P(X = 2)$. (b) $P(X \geq 1)$.
17. Approximate $\text{Binomial}(200, 0.01)$ with Poisson. Find $P(X = 0)$ and $P(X = 2)$.
18. Cable defects: 3 per 10 m. (a) $P(X = 0)$ for 10 m. (b) Expected defects per 5 m. (c) $P(X = 0)$ for 5 m.
19. NLP: word “serendipity” averages 0.3 per 1000 words. $P(\text{at least one in 1000 words})$?
20. Verify: $\text{Binomial}(50, 0.06) \approx \text{Poisson}(3)$ by comparing $P(X = 0)$, $P(X = 1)$, $P(X = 2)$.
21. Show $\sum_{k=0}^{\infty} P(X = k) = 1$ using $e^{\lambda} = \sum_{k=0}^{\infty} \lambda^k / k!$.
22. Call centre: 12 calls/hour. (a) Expected calls in 15 minutes. (b) $P(X = 0)$ in 5 minutes. (c) What distribution models the time between calls?

Part C: Continuous Distributions (8 problems)

For continuous X : $P(a \leq X \leq b) = \int_a^b f(x) dx$ where f is the PDF.

Uniform(a, b): $f(x) = \frac{1}{b-a}$; mean = $\frac{a+b}{2}$; variance = $\frac{(b-a)^2}{12}$.

Exponential(λ): $f(x) = \lambda e^{-\lambda x}$; mean = $\frac{1}{\lambda}$; $P(X > x) = e^{-\lambda x}$; **memoryless**.

23. Random number $X \sim \text{Uniform}(0, 10)$. (a) Sketch f . (b) $P(3 \leq X \leq 7)$. (c) $P(X > 8)$. (d) Mean and variance.
24. Bus arrives uniformly between 9:00–9:10. (a) $P(\text{wait} > 7 \text{ min})$. (b) Expected wait.
25. Customer calls: rate 6/hour ($\lambda = 0.1/\text{min}$). (a) Expected wait between calls (min). (b) $P(\text{wait} > 15 \text{ min})$. (c) $P(\text{wait} \leq 5 \text{ min})$.
26. Verify the memoryless property: $P(X > s + t \mid X > s) = P(X > t)$ using $P(X > x) = e^{-\lambda x}$.

27. Website: 4 visits/minute. (a) Distribution of time between visits. (b) $P(\text{next visit} > 1 \text{ min})$. (c) Expected time between visits.
28. For the exponential distribution, $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Verify $\int_0^\infty f(x) dx = 1$.
29. Which distribution (Uniform, Exponential, or Poisson) is most appropriate for each scenario? (a) Number of goals in a 90-minute match. (b) Time between phone calls. (c) Value of a randomly selected measurement between two fixed limits. (d) Number of misprints on a page.
30. A component fails with $\text{Exp}(\lambda = 0.01)$ per hour. (a) Mean lifetime. (b) $P(\text{lasts} > 100 \text{ h})$. (c) $P(\text{fails within } 50 \text{ h})$.

Part D: The Normal Distribution (8 problems)

$$X \sim N(\mu, \sigma^2): f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Standardise: $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.

68-95-99.7 rule: 68% within 1σ ; 95% within 2σ ; 99.7% within 3σ .

31. Male heights: $N(178, 64)$ (mean 178 cm, $\sigma = 8$ cm). (a) 68% interval. (b) 95% interval. (c) Fraction taller than 194 cm. (d) z -score for 162 cm.
32. Exam scores: $N(65, 100)$ ($\sigma = 10$). (a) Score 1.5 SD above mean. (b) Fraction between 55 and 75. (c) Threshold for top 2.5%.
33. Bolts: $N(10, 0.01)$ mm ($\sigma = 0.1$). Acceptable: 9.8–10.2 mm. (a) Express limits as z -scores. (b) Fraction acceptable. (c) How to reduce rejection rate?
34. $X \sim N(50, 25)$ ($\sigma = 5$). Using $P(Z < 1) \approx 0.841$, $P(Z < 2) \approx 0.977$: (a) $P(X < 55)$. (b) $P(45 < X < 60)$. (c) 97.5th percentile.
35. **MSE and Gaussian MLE.** If prediction errors $\sim N(0, \sigma^2)$, the MLE of σ^2 is $\hat{\sigma}^2 = \frac{1}{n} \sum \epsilon_i^2$. Explain why minimising MSE is equivalent to MLE under Gaussian noise.
36. Log-normal: if $\ln X \sim N(2, 1)$, what is the median of X ? Why is the distribution right-skewed?
37. True or false: (a) The Normal distribution is symmetric about its mean. (b) $P(X = 3) > 0$ for $X \sim N(0, 1)$. (c) The standard deviation can be negative. (d) Approximately 95% of data lies within 2σ of the mean.
38. The sum of two independent normals $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ is $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Verify the variances add (not standard deviations) with a specific example.

Part E: Central Limit Theorem (6 problems)

If $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 , then for large n :

$$\bar{X} = \frac{1}{n} \sum X_i \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

39. Die rolled 100 times; single roll: mean 3.5, variance $35/12$. (a) Approximate distribution of total score S . (b) $P(S > 370)$. (c) $P(3.3 < \bar{X} < 3.7)$.
40. Binomial(100, 0.4) approximated by normal. (a) Parameters. (b) $P(X \leq 45)$. (c) When is the approximation most accurate?
41. 500 insurance claims per day; claim time: mean 6 min, SD 3 min. (a) Distribution of total time. (b) $P(\text{total} > 3100 \text{ min})$.
42. **Ensemble learning.** If each of n independent models has error variance σ^2 , the average model has variance σ^2/n . Explain why this justifies using many trees in a Random Forest.
43. Give an example where the CLT does *not* apply because of dependency.
44. Connections: (a) Binomial and Bernoulli. (b) Poisson and Markov chains (memoryless property). (c) Normal and MSE minimisation. (d) CLT and the sample mean from ws06a.